



Automated cement fragment image segmentation and distribution estimation via a holistically-nested convolutional network and morphological analysis

Huaian Chen ^a, Yi Jin ^{a,*}, Guiqiang Li ^{a,*}, Biao Chu ^b

^a School of Engineering Science, University of Science and Technology of China, Hefei, People's Republic of China

^b Hefei Cement Research & Design Institute, Hefei, People's Republic of China

ARTICLE INFO

Article history:

Received 30 June 2017

Received in revised form 5 February 2018

Accepted 4 August 2018

Available online 06 August 2018

Keywords:

HCN

Segmentation

Morphological analysis

Cement fragment distribution

ABSTRACT

The distance between two rollers in a ball grinding mill is determined by the size of cement fragments. As such, automated detection of fragment size distribution is of great importance to the cement production industry. Therefore, we propose a holistically-nested convolutional network (HCN) and corresponding morphological analysis to estimate cement fragment distributions. This procedure can be divided into three stages. First, a cement fragment image training set was input to the HCN, helping the algorithm distinguish between fragments and the background. A series of morphological operations including morphological clean and opening operation were then employed to improve segmentation performance and the accuracy of fragment size calculation. Finally, the 'open' operation was employed to obtain the fragment distribution from a segmented image. Experimental results demonstrated that the segmentation and size calculation achieved using our algorithm were superior to those using comparable conventional techniques. The precision, F_1 -measure, PRI and VI using proposed method are improved at least 1.3%, 3.3%, 9.5% and 36.5% respectively.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

Grinding processes used in ball mills are the most effective way to produce cement, which has become indispensable in modern construction, on a large scale. The distance between any two rollers in the grinding apparatus, a critical operational parameter, is determined by the size of cement fragments. As such, cement fragment distribution has a significant effect on grinding efficiency. Currently, cement fragment distributions are measured via manual sampling, which is time consuming and does not provide real-time feedback for automated control purposes. Segmentation of cement fragment image is the key to automatically estimate cement fragment distributions. However, automated cement segmentation images have few complications: (1) cement fragments are often in contact with each other and overlap, (2) stains are often present in the images, and (3) cement fragments vary widely in size, shape, appearance, and texture.

In this study, we propose a novel hybrid strategy to overcome these challenges. We first review conventional segmentation methods and introduce their application to actual object segmentation. Then, we

discuss the similarity of rock fragment and cement fragment image segmentation. Finally, the motivation for developing a novel approach which overcomes current limitations is presented.

1.1. Related work

Segmentation is a common image processing technique used in a wide variety of fields. Various algorithms have been proposed, studied, and applied to increasingly diverse applications [1–4]. For example, Otsu [5] presented a nonparametric and unsupervised method of automatic threshold selection for picture segmentation, in which an optimal threshold was selected by discriminant criteria. Juraj Horváth [6] applied fuzzy c-means clustering to classify pixels into appropriate segments. Beucher [7] established an intuitive definition of the watershed transform and applied it to image segmentation. Chen et al. [8] proposed an application of Otsu's method to segment nuclei in a background and deployed a watershed technique to further separate overlapping nuclei. Salinas et al. [9] developed a watershed-based algorithm capable of segmenting rock fragment images automatically.

Despite these developments, there are currently few applications of traditional segmentation methods to cement. This is because analysis of such images becomes significantly more difficult for the three reasons. First, fragments are in contact with each other and often overlap. Second, stains are commonly present in these images. Finally, the size,

* Corresponding authors.

E-mail addresses: jinyi08@ustc.edu.cn (Y. Jin), ligq@mail.ustc.edu.cn (G. Li), chubiao@mail.ustc.edu.cn (B. Chu).

shape, appearance, and texture of fragments vary significantly. In this study, we found the segmentation of cement fragments to be highly similar to segmentation of rock fragments. Each includes challenges such as object overlap and texture noise. All these methods [9–11] employed in rock fragment segmentation can be used to segment cement fragments, but their segmentation accuracy is not enough to calculate the cement fragment size accurately.

Recently, there has been a growing interest in the application of convolutional neural networks for the analysis of image segmentation. In [12], deep convolutional neural networks (DCNNs) were applied to image segmentation, which achieved a great performance in high level vision tasks. In [13], conditional random fields (CRFs) were used to learn CNN features, in order to produce a more accurate segmentation. In [14], fully convolutional networks (FCNs) were proposed to segment objects against a background. However, these methods perform poor segmentation in case of complicated images of cement fragment. To attain good performance, modification and post-processing is required in these methods for effective application to cement fragment fields. Based on this, we propose a holistically-nested convolutional network (HCN), combined with morphological analysis, to process cement fragment images. The HCN was developed from the literature [15] by modifying the loss function. In this study, we apply it to the cement fragment segmentation because the pixels distribution in cement fragment segmentation is similar to the image edge detection. One type of the pixels (cement pixels or backgrounds pixels) is account for more than 90%. Additional, morphological operations are also employed to improve the performance of segmentation and calculate the cement fragment distributions. Experimental results demonstrate that the proposed method can effectively suppress noise, avoid over segmentation and calculate cement fragment size accurately.

1.2. Contributions of our work

In this study, we applied the HCN to cement segmentation because the pixel distribution in cement fragments is similar to edge detection. A single pixel type (cement pixels or backgrounds pixels) accounts for more than 90% of the image. The primary contributions of this work include three points.

First, a dataset was built for cement fragment segmentation and a holistically-nested convolutional network developed from literature [15], which alters the loss function, was applied to the segmentation of cement fragments. Morphological operations were introduced in the post-processing step to improve segmentation performance and calculation accuracy for cement fragment distributions.

Second, cement fragment size detection, based on the morphological 'open' operation, was employed to obtain the cement fragment distribution. We first increased the size of this morphological structural element. The cement fragment was filtered when the radius of maximum circle contained in the cement fragment was smaller than the morphological structural element's radius. Then, the corresponding unfiltered cement fragment pixel values were then added in each opening operation. Each value represented the cumulative area of cement fragments larger than a certain size, which represented by the size of morphological structural element.

Finally, extensive experimental results (and comparisons with common methods) demonstrated that the proposed method can effectively suppress noise, avoid over segmentation, and accurately calculate cement fragment size. As such, the proposed method is a promising new tool for segmentation of industrial cement images.

In the following sections, we develop and demonstrate this technique. Section 2 presents our methodology in detail, while Section 3 describes the experimental setup and several comparisons made with other methods. Section 4 provides a discussion and section 5 concludes the paper.

2. Methodology

We propose an efficient strategy for automatic segmentation of cement fragment images and calculate cement fragment distributions. An overview of the proposed method is shown in Fig. 1. First, the HCN, a convolutional neural network developed using previous studies [15] was trained to efficiently segment cement fragments from the background. This process is explained in detail in the following sections.

A series of morphological operations were then introduced during post-processing to improve segmentation performance and the accuracy of cement fragment size calculations. Finally, cement fragment size detection, based on the morphological 'open' operation, was employed to obtain the fragment distribution and the cumulative area distribution.

2.1. Holistically-nested convolutional network

The HCN was developed from a trimmed VGG net which is produced by cutting the last stage, including all fully-connected layers and the fifth pooling layer. As shown in Fig. 2, an additional deep supervision step was added to the trimmed VGG net. Additional deep supervision was established by connecting the side output layer to the last convolutional layer in each stage, conv1_2, conv2_2, conv3_3, conv4_3, and conv5_3, respectively. The side output layer was implemented as a convolutional layer with a kernel size of 1 and a single output. In addition, the HCN added a fusion layer to the network which can link all side output layer predictions and learn fusion weights during training. As a result, the overall loss function L can be written as:

$$L(I, G, W, w) = L_{side}(I, G, W, w) + L_{fuse}(I, G, W, w) \quad (1)$$

where L_{fuse} denotes the fusion layer loss function, L_{side} denotes the side output layer loss function, I denotes the raw input image, and G denotes the corresponding ground truth binary segmentation map. The term $w = (w(1), \dots, w(M))$ denotes the corresponding weights for each side output layer. W denotes the collection of all other network layer parameters.

The loss function for this HCN has been modified from the network in [15] because most of the pixels in a cement image are positive (fragment pixels), in contrast to edge detection where 90% of pixels are negative (background pixels). The class-balancing weight β_j can then be described as:

$$\beta_j = \frac{I_+}{I} \quad (2)$$

where the index j is over spatial image dimensions of I . The terms I_+ and I denote the number of cement fragment (positive) pixels and the total number of pixels, respectively.

2.1.1. Training workflow with HCN

The HCN was trained using the image-to-image approach. When an image was input to the HCN, as shown in Fig. 1, the result is five side output maps and one fusion output map. The five side-output layer loss, fusion layer loss, and overall loss were then calculated. Afterwards, a back-propagation strategy was used to adjust network weights according to these losses (L_{side}, L_{fuse}, L). This produced the HCN model used for cement fragment segmentation.

2.2. Post-processing

A series of morphological operations were employed to improve segmentation performance and fragment size calculation accuracy. First, a large threshold was selected to produce a binary image. This threshold degrades segmentation performance, but improves size calculation accuracy by reducing the probability of fragments overlapping

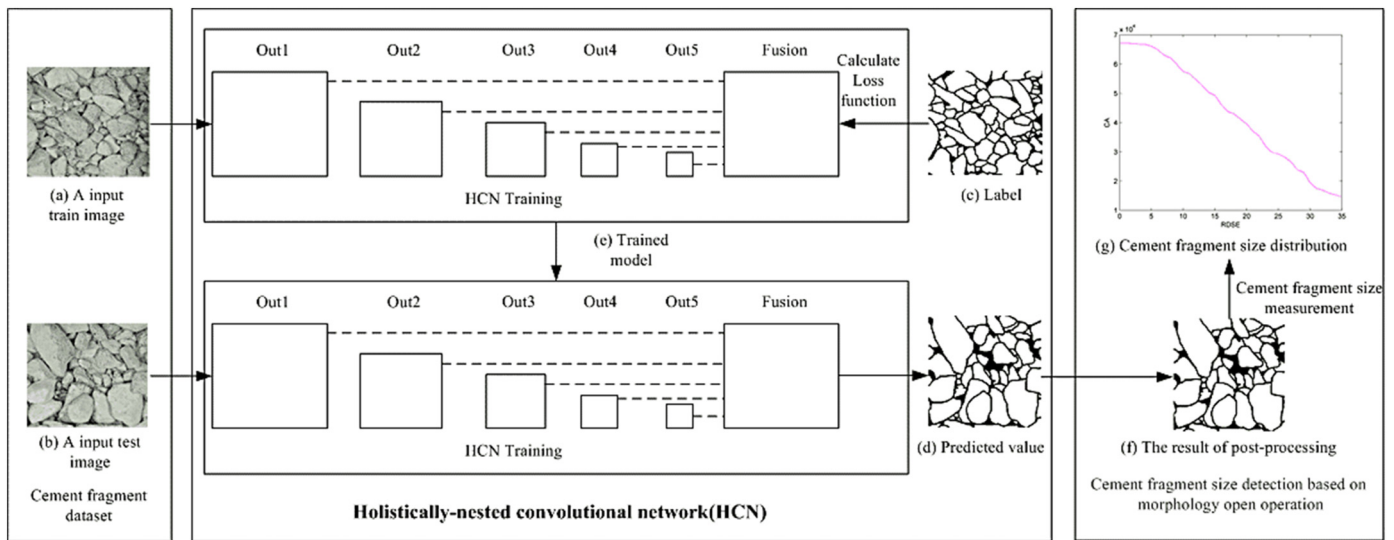


Fig. 1. An overview of the proposed method.

each other. In this study, we find that the threshold between 0.55 and 0.65 will get a better result. Our threshold was set to be 0.6. Morphological cleaning and opening operations were then employed to eliminate isolated pixels and small fragments, due to noise and texture. Finally, the dilation operation was used because larger thresholds cause the output image to be thinner than the ground truth.

2.3. Cement fragment size measurement

After HCN-based segmentation and post-processing, the ‘open’ operation was employed to obtain fragment distributions from cement images. In this study, the actual area was not calculated but rather a ratio between fragment and background pixels. Fragment gray-scale values were set to 1 and background gray-scale values were set to 0. The cement fragment area can then be represented by the sum of segmentation pixel values. If the camera parameters such as angle and distance were fixed, the sum of segmentation pixels will correspond to the physical area through calibration. Based on this, the cement fragment size distribution can be calculated using the following steps:

- (1) The radius of a disk-shaped structural element (RDSE) is gradually increased using the morphological opening operation.

- (2) The corresponding unfiltered cement fragments pixel values are added in each opening operation. These calculated values represent the cumulative area (CA) of cement fragments larger than the disk structural element size.
- (3) A cement fragment distribution map is produced by plotting RDSE on the abscissa and CA on the ordinate.

We select a disk-shaped structural element because the diameter of the maximum circle contained in the cement fragment can express the width of cement fragment bounding rectangle in a better way as compared to any other structure element. The cement fragment was filtered when the radius of maximum circle contained in the cement fragment was smaller than the morphological structural element's radius. When the ‘open’ operation is employed to obtain the fragment distribution, the diameter of the maximum circle contained in the cement fragment can be approximately regarded as the width of cement fragment bounding rectangle. Few adjacent cement fragments are difficult to separate because they are often in contact with each other or overlap and therefore, regarded as one. In comparison to diameter of maximum circle contained in fragment, the equivalent area diameter used in [16] is

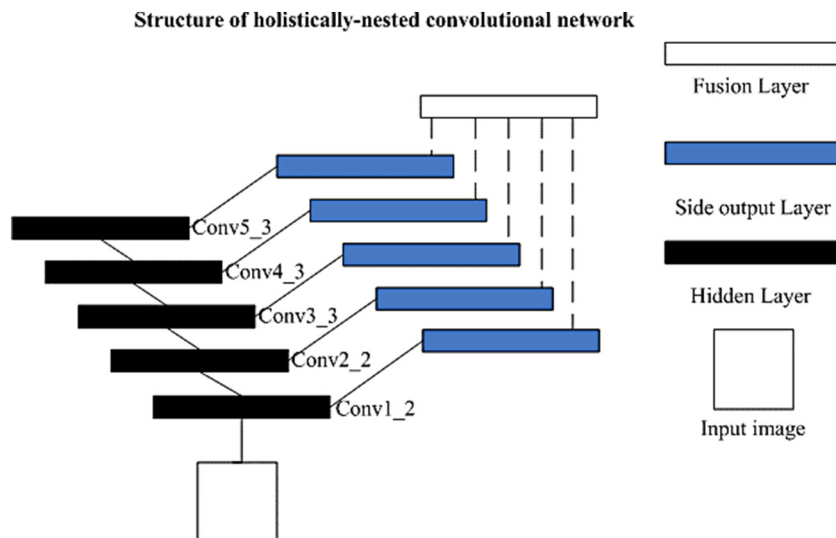


Fig. 2. Structure of the proposed holistically-nested convolutional network.

always large and the width of bounding rectangle used in [17] includes adjacent fragments too. Hence, the diameter of maximum circle contained in the cement fragment can express the fragment size in a better way.

3. Experimental configuration

In this section, we discuss the implementation procedure and introduce the performance metrics used in present work. HCN framework was implemented using the publicly available CAFFE Library. While, the cement fragment size was calculated using MATLAB R2013a

3.1. Dataset description

A series of experiments were conducted in which a camera system was used to acquire cement fragment images of various sizes. These images were collected from the cement research and design institute in Hefei, China. The pixels of images with missing cement fragment or with cement fragment present in lower layers are defined as background pixels to ensure ground truth. The presence of cement fragment in lower layers leads to unrecognizability of fragment shape. Therefore, they are also defined as background pixels. 150 original images were used to train the model. Data augmentation [15, 18] including resizing, flipping, rotation and elastic deformation was performed on image dataset. The sample images were resized to different sizes of 350×350 , 300×300 , 250×250 , and 200×200 . Further, these images were flipped horizontally or vertically and rotated in the range of -180 to $+180$ degrees randomly. The purpose of expanding data was to improve the accuracy and multi-scale adaptability of our model. An additional 50 images were used for evaluation. These were classified into five groups, each containing ten images of size 400×400 , 350×350 , 300×300 , 250×250 , and 200×200 . Fig. 3(a) shows an input image fragment from the test set with a size of 300×300 . Fig. 3(b) shows the corresponding ground truth.

3.2. Segmentation performance metrics

Multiple performance metrics have been used in the literature to measure image segmentation accuracy [19, 20]. In this paper, we first use basic evaluation criterion: recall, precision, and F_1 -measure. Additionally, the variation of information (VOI) and probabilistic rand index (PRI) were employed to evaluate segmentation performance.

3.2.1. Precision, recall, and F_1 -measure

For the proposed method, precision denotes the fraction of segmented cement fragments that are present in the ground truth image. Recall denotes the fraction of the fragments in the ground truth image detected by our method. F_1 -measure is the harmonic mean of the precision and recall. Precision (P), Recall (R), and F_1 -measure (F) are given as:

$$P = \frac{TP}{TP + FP} \quad (3)$$

$$R = \frac{TP}{TP + FN} \quad (4)$$

$$F = \frac{2 \times P \times R}{P + R} \quad (5)$$

where TP (true positive) represents cement fragments in the ground truth image which were correctly detected by the proposed method. FP (false positive) refers to fragment regions detected by our method which were not present in the ground truth. FN (false negative) denotes cement fragments present in the ground truth which were not detected using our method.

3.2.2. Variation of information

Melia proposed a method using variation of information to give an axiomatic characterization of criteria for comparing clustering [21]. Arbelaez et al. applied this algorithm to measure the distance between two segmentation results in terms of their average conditional entropy [20], which can be written as:

$$VI(S, G) = H(S) + H(G) - 2I(S, G) \quad (6)$$

where VI denotes the variation of information between S and G. H and I represent the entropy and mutual information between the segmentation result S and the ground truth G, respectively.

3.2.3. Rand index

The rand index is a measure of the similarity between two data clustering results. It has been used in previous studies to evaluate image segmentation [20, 22, 23]. Given a segmentation result S and a set of

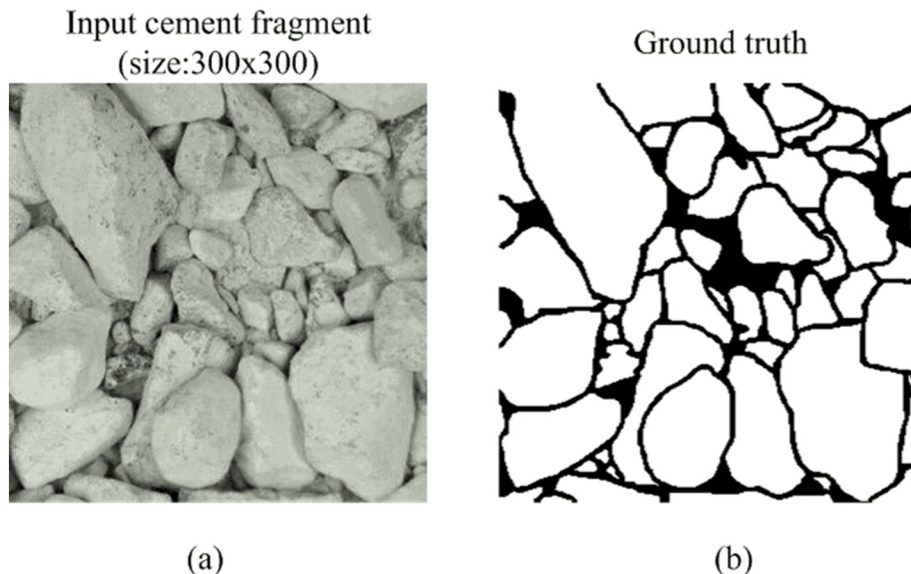


Fig. 3. The a) input cement fragment image and b) ground truth.

ground truth data $\{G_k\}$, the probabilistic rand index (PRI) is defined as:

$$PRI(S, \{G_k\}) = \frac{1}{T} \sum [c_{ij}p_{ij} + (1-c_{ij})(1-p_{ij})] \quad (7)$$

where c_{ij} is the event that pixel i and j have the same label and p_{ij} its probability. T denotes the total number of pixel pairs. k denotes the number of ground truth images.

3.3. Parameters

- (1) HCN: The momentum was set to 0.9 and the weight decay to 0.0002. Initialization of the nest filters was 0 and initialization of the fusion layer weight was 0.2. The HCN will not function satisfactorily if the learning rate is larger than $1E-7$. Therefore, the learning rate was set to $1E-8$. In addition, to obtain a better performance model, the learning rate was reduced by 1/10 every 10,000 training iterations.
- (2) Post-processing: For each candidate cement fragment cluster, the area of each cement fragment region was obtained by morphological opening operation and the structure element is selected as a disk. In this paper, the radius was set to 3 pixels empirically. The dilation operation was used with a diamond structure element and an empirically-selected radius of 1 pixel.

3.4. Segmentation results and comparisons

Five side output maps and one fusion output map are shown in Fig. 4. It is evident the side output maps shown in (a) through (e) are different scale features. The fusion output map shown in (f) combines these different scale features into an accurate segmentation map. Fig. 5

shows the post-processing results. Fig. 5(a) is the resulting HCN segmentation. The cement fragments in Fig. 5(b) were always thinner than the ground truth in Fig. 3(b). In contrast, Fig. 5(c) is a better representation of the ground truth.

A series of experiments were conducted to compare the proposed method with three commonly-used conventional techniques in the same test set. These included FCM-based [6] and Watershed-based [7] methods, as well as the technique proposed by Otsu [5].

Otsu's method [5] automatically calculates the most appropriate threshold for image segmentation. Our cement images contain two pixel classes: fragment pixels and background pixels. Otsu's method calculates the optimal threshold to maximize inter-class variance. Fuzzy c-means [6] is a clustering algorithm based on partitions. FCM was employed to achieve maximal similarity between objects divided into the same cluster (cement fragment or background), while minimizing the similarity between different clusters. Watershed-based [7] algorithms regard cement fragment images as topological geomorphology. The gray values of each pixel in the cement fragment image indicate the altitude at that point. Each local minimum and its affected area is called a basin and the boundaries of the basin form a watershed.

Comparison results are shown in Table 1. The reported value is the average of each measured performance in the test set. It is evident the proposed method (HCN) outperformed these conventional techniques across all metrics. As shown in Table 1, the similarity between ground truth and segmentation was maximal with the HCN method, whose PRI was 0.84. The distance between ground truth and segmentation result was minimal using the HCN method, with a VI of 0.74. We also found the precision to be 1.3% higher than for other methods, while the F_1 -measure improved at least 3.3%. In addition, recall for the HCN method was close to that of the watershed-based method, while other metrics were significantly higher.

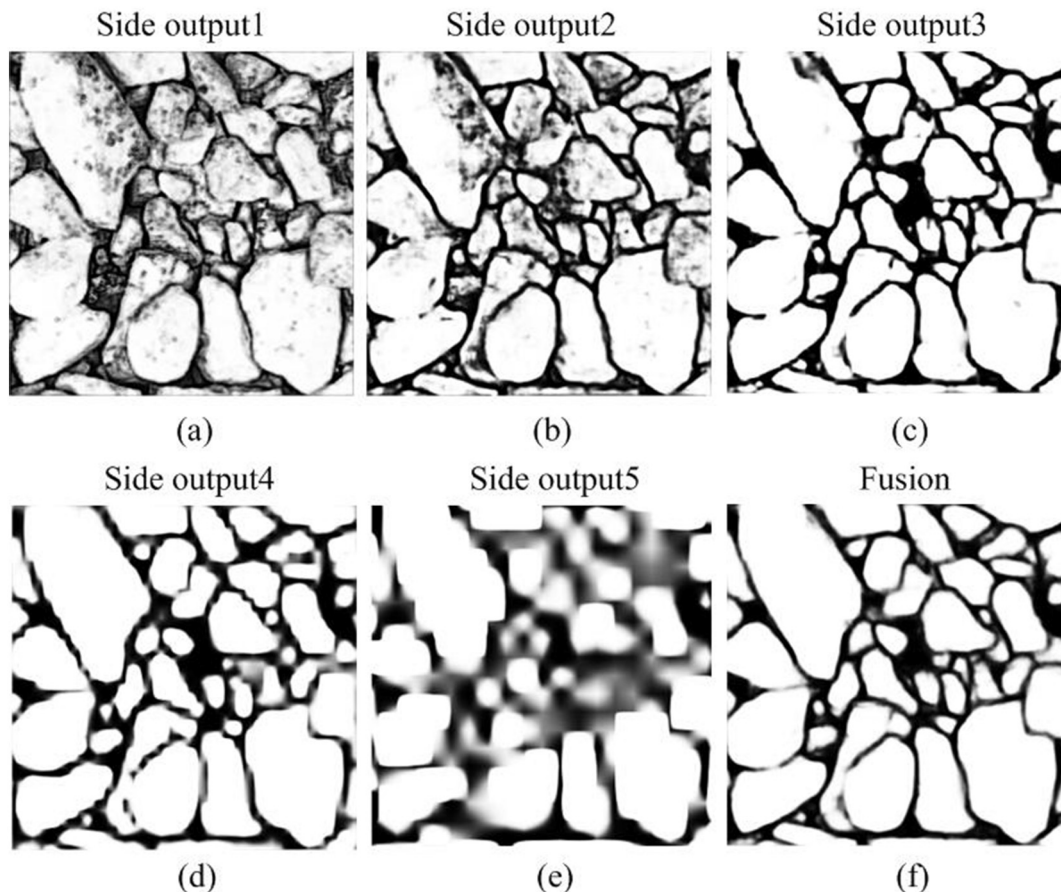


Fig. 4. Side output maps a)-e) and the fusion output map f).

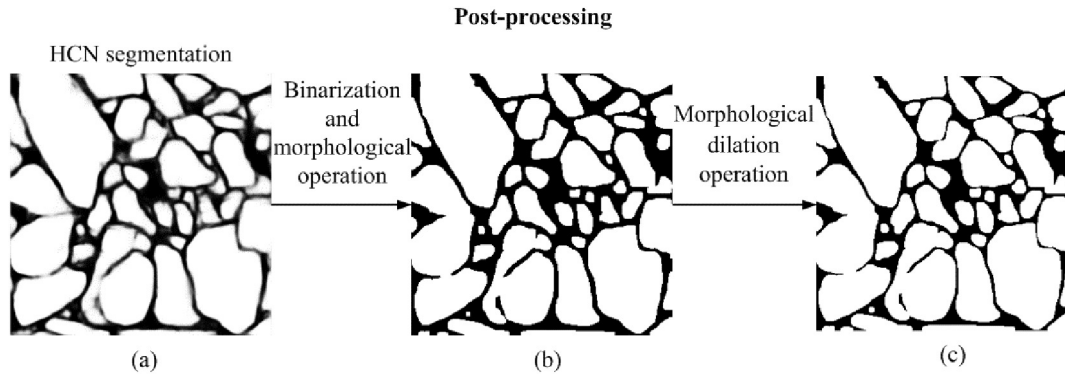


Fig. 5. Post-processed segmentation results.

Segmentation results are shown in Fig. 6. The proposed method was able to accurately extract cement fragments. In contrast, we also found that FCM-based algorithms tended to cause less segmentation while watershed-based methods often caused over-segmentation. The segmentation obtained using Otsu's algorithm has several holes.

3.5. Calculation of cement fragment size

After HCN segmentation and image post-processing, the cement fragment size can be calculated using the method described in Section 2.3. A first order polynomial was used to fit the calculated cement fragment size with the ground truth data. The calculated coefficients and R-square values of the polynomial regression are shown in Table 2. R-square value calculated from the data obtained from proposed method is higher than the values for other methods, which confirms its better performance. A cement fragment cumulative area distribution map is shown in Fig. 7. The half width of each fragment bounding rectangle and the cumulative area of the cement fragments are used to plot the ground truth curve (red line). In this case, the width of each fragment bounding rectangle is directly used because the boundary of cement fragment in the ground truth image is segmented accurately. We find the pink line, calculated from the HCN segmentation, is nearest to the red line. This suggests the segmentation obtained using our method is a more physical representation of the image. The areas in each disk structural element can be also calculated from the differential coefficient of the cement fragment cumulative area (CA). As shown in Fig. 8, ACF denotes the area of cement fragments in each disk structural element. In other words, it denotes the area of a certain size cement fragment represented by a certain radius disk structural element.

4. Discussion

Distinguishing between each cement fragment is very difficult, especially when texture and overlap are considered. As such, an HCN was utilized to assist in making this distinction. The HCN has two important properties [24–26]. Firstly, it encodes high-level domain-specific knowledge into the segmentation strategy by acquiring training data and combining five side output maps into a single fusion output map. This produces sufficient multi-scale information. Secondly, the HCN uses appropriate amounts of contextual information by optimizing the

weights of network filters through the gradient descent learning process.

As a result, the cement fragment segmentation problem is transformed into a pixel classification issue. As shown in Fig. 6, HCN segmentation is more accurate than FCM-based segmentation. A slight contrast exists between cement fragments and the background. Moreover, the cement fragments are all very similar. This causes the FCM-based algorithm to produce less segmentation. As shown in Fig. 6(b) and Fig. 3(a), the segmentation obtained using Otsu's algorithm has many holes in areas where the texture is obvious. From Fig. 6(c), we observe the watershed-based algorithm often struggles to distinguish between cement fragments and the background, which can result in over-segmentation.

Performance metric including precision, recall, and F₁-measure are very large compared to those evaluated in [26, 27]. This is because most pixels in the cement fragment image are positive (cement fragment) pixels. These positive pixels account for more than 90% of the image, which results in a high TP and a low FP. Therefore, the precision will then be larger than in other studies. In some cases, The FN can be a relatively large number. For example, the recall calculated using the FCM-based method was 0.7434. But in most cases, Unlike TP, the FN achieved using most methods is fairly low.

The goal of this study was to segment a variety of cement fragments and calculate their distribution. Fig. 7 shows that cement fragments calculated using HCN segmentation are nearest to the red line calculated from the ground truth. The proposed method can improve the accuracy of fragment size calculation for two important reasons. Firstly, the HCN model combined with morphological analysis segment the cement fragment image more accurately, and there is a strong correlation between segmentation performance and accuracy of fragment size calculation. Secondly, the diameter of the maximum circle contained in the cement fragment is taken as fragment width that express the fragment width in a better way as discussed in Section 2.3.

The WipFrag sizing system was designed to measure fragmentation online, over a conveyor belt [28–30]. It was developed by Maerz and Franklin at the University of Waterloo and has been in widespread use for many years. It is the automated standard for estimating rock fragment size using image segmentation. Rock pile sampling and photography strategy in WipFrag sizing system can be also used in our work.

The proposed HCN model can be used for cement fragment segmentation in an online sieve monitoring system. For this purpose, the model should be trained offline using real-time samples. The HCN model could provide additional real-time features. Further testing of camera parameters (angle and distance) and study of environmental interference is also required before applying this method to online sieve monitoring industrial processes. This work will be performed in our future study.

HCN model combined with morphological analysis can also be applied for particle size detection of rock fragment and coal piles during mineral processing and coal-mining respectively. It can also be used in the cell image analysis such as segmentation of nuclei in pathological

Table 1
A comparison of segmentation performance results.

Method	Precision	Recall	F ₁ -measure	PRI	VI
FCM	0.9449	0.7434	0.8321	0.64	1.35
Otsu	0.9224	0.8895	0.9057	0.75	1.09
Watershed	0.9115	0.9245	0.9180	0.76	1.01
HCN (proposed)	0.9572	0.9414	0.9492	0.84	0.74

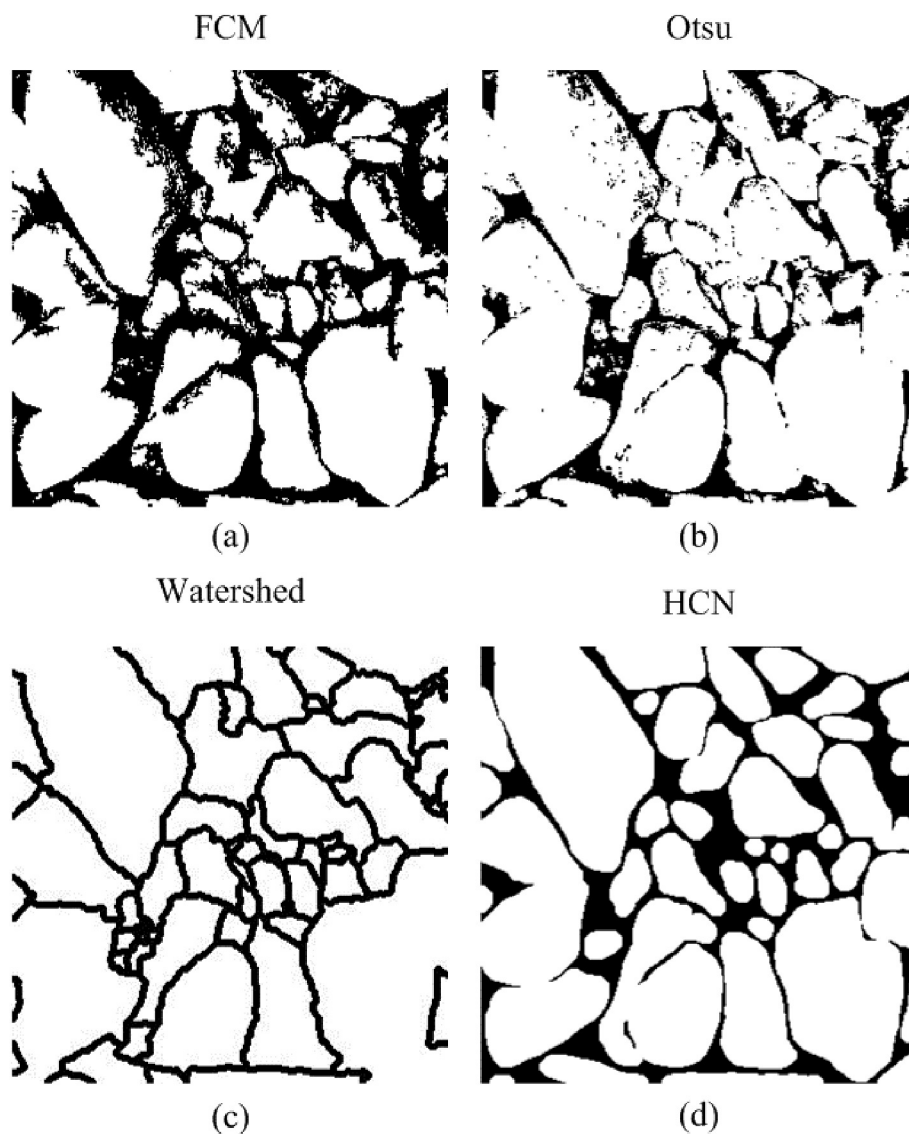


Fig. 6. Test image segmentation results achieved using an a) FCM-based method, b) Otsu's method, c) a watershed-based method, and d) the proposed method.

images. Fine turning is required when the HCN model is used in other images directly.

5. Conclusion

We proposed an automatic image processing technique to accurately segment cement fragment images and calculate fragment size. We first employed an HCN and trained it to distinguish cement fragments from the background. A series of morphological operations were then applied to improve segmentation performance and size calculation accuracy. Finally, the morphological 'open' operation was employed to obtain the cement fragment distribution. The proposed

Table 2
The coefficients and R-squared values of the calculated fragments size compared to the ground truth data.

Method	Coefficients		R-square
	Coefficient1	Coefficient2	
FCM	1.010	24,857.2	0.9047
Otsu	0.9171	10,131.2	0.9796
Watershed	1.135	−8649.4	0.9930
HCN	1.085	−3468.1	0.9983

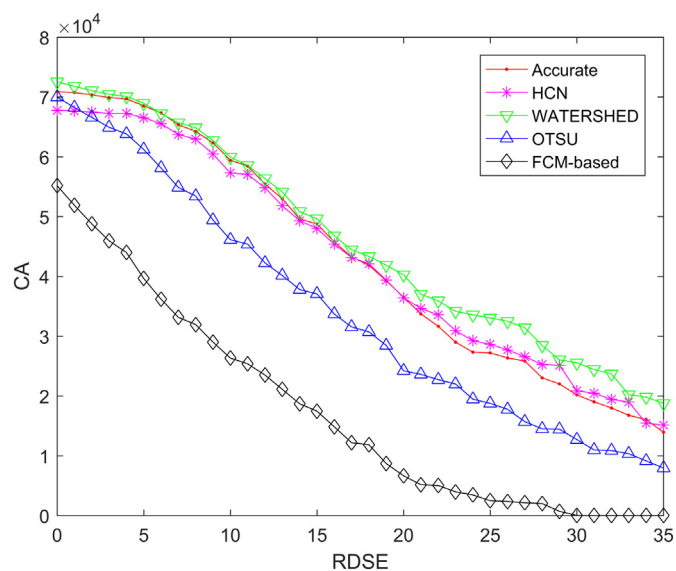


Fig. 7. The cement fragment cumulative area distribution map.

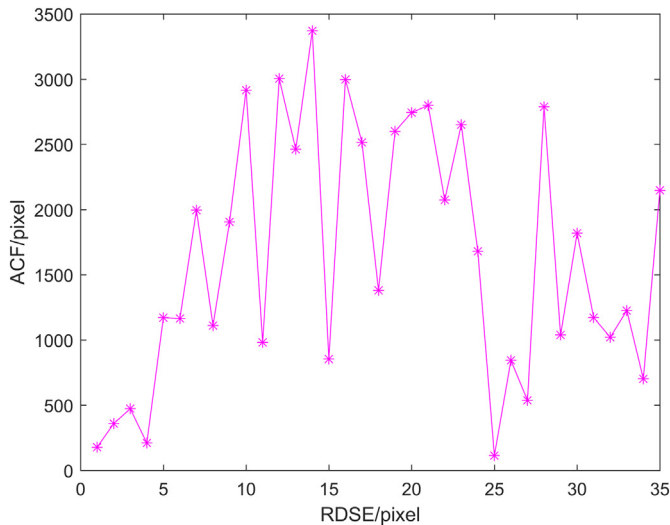


Fig. 8. Cement fragment area maps for each disk structural element.

method is a general approach which could be adapted to several other particle segmentations, such as rock fragments. Experimental results in Table 1 demonstrated that the segmentation performance and size calculation accuracy achieved using our method were superior to those using conventional techniques.

Acknowledgment

This research was supported by the National Science Foundation of China (Grant Nos. 51605464), National Basic Research Program of China (973 Program) (2014CB049500).

References

- [1] N.R. Pal, S.K. Pal, A review on image segmentation techniques, *Pattern Recogn.* 26 (9) (1993) 1277–1294.
- [2] N.M. Zaitoun, M.J. Aqel, Survey on image segmentation techniques, *Procedia Computer Sci.* 65 (2015) 797–806.
- [3] Z. Zhang, J. Yang, L. Ding, Y. Zhao, Estimation of coal particle size distribution by image segmentation, *Int. J. Min. Sci. Technol.* 22 (5) (2012) 739–744.
- [4] N. Sadr-Kazemi, J.J. Cilliers, An image processing algorithm for measurement of flotation froth bubble size and shape distributions, *Miner. Eng.* 10 (10) (1997) 1075–1083.
- [5] N. Otsu, A threshold selection method from gray-level histograms, *IEEE Trans. Syst. Man Cybernetics* 9 (1) (1979) 62–66.
- [6] J. Horvath, Image segmentation using fuzzy c-means, *Symposium on Applied Machine Intelligence*, 2006.
- [7] S. Beucher, Watersheds of functions and picture segmentation, *Acoustics, Speech, and Signal Processing, IEEE International Conference on ICASSP'82*, vol. 7, 1982, pp. 1928–1931.
- [8] X. Chen, X. Zhou, S.T. Wong, Automated segmentation, classification, and tracking of cancer cell nuclei in time-lapse microscopy, *IEEE Trans. Biomed. Eng.* 53 (4) (2006) 762–766.
- [9] R.A. Salinas, U. Raff, C. Farfan, Automated estimation of rock fragment distributions using computer vision and its application in mining, *IEE Proc. Vision Image Signal Proc.* 152 (1) (2005) 1–8.
- [10] Q.J. Qin, X.J. Qiang, Y. Liu, J. Yang, Rock image pore identification based on fuzzy C-means clustering and neural networks, *Applied Mechanics and Materials*, vol. 571, Trans Tech Publications 2014, pp. 803–806.
- [11] C. Farfán, R.A. Salinas, G. Cifuentes, Rock Segmentation and Measures on Gray Level Images Using Watershed for Sizing Distribution in Particle Systems, *Electrical Engineering Department, University of Santiago, Chile (USACH)-Metallurgical Engineering Department*, 2006.
- [12] L.C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, A.L. Yuille, Semantic image segmentation with deep convolutional nets and fully connected crfsarXiv preprint arXiv:1412.70622014.
- [13] F. Liu, G. Lin, C. Shen, CRF learning with CNN features for image segmentation, *Pattern Recogn.* 48 (10) (2015) 2983–2992.
- [14] J. Long, E. Shelhamer, T. Darrell, Fully convolutional networks for semantic segmentation, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* 2015, pp. 3431–3440.
- [15] S. Xie, Z. Tu, Holistically-nested edge detection, *Proceedings of the IEEE International Conference on Computer Vision* 2015, pp. 1395–1403.
- [16] Zelin Zhang, et al., Multi-scale image segmentation of coal piles on a belt based on the Hessian matrix, *Particuology* 5 (11) (2013) 549–555.
- [17] Zelin Zhang, et al., An improved estimation of coal particle mass using image analysis, *Powder Technol.* 229 (2012) 178–184.
- [18] Olaf Ronneberger, Philipp Fischer, Thomas Brox, U-net: convolutional networks for biomedical image segmentation, *International Conference on Medical Image Computing and Computer-Assisted Intervention* 2015, pp. 234–241 (Springer, Cham).
- [19] M. Beneš, B. Zitova, Performance evaluation of image segmentation algorithms on microscopic image data, *J. Microscopy* 257 (1) (2015) 65–85.
- [20] P. Arbelaez, M. Maire, C. Fowlkes, J. Malik, Contour detection and hierarchical image segmentation, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33 (5), 2011, pp. 898–916.
- [21] M. Meilă, Comparing clusterings: an axiomatic view, *Proceedings of the 22nd International Conference on Machine Learning, ACM* 2005, pp. 577–584.
- [22] R. Unnikrishnan, C. Pantofaru, M. Hebert, Toward objective evaluation of image segmentation algorithms, *IEEE Trans. Pattern Anal. Mach. Intell.* (2007) 29(6).
- [23] A.Y. Yang, J. Wright, Y. Ma, S.S. Sastry, Unsupervised segmentation of natural images via lossy data compression, *Comput. Vis. Image Underst.* 110 (2) (2008) 212–225.
- [24] Matthew D. Zeiler, Rob Fergus, Visualizing and understanding convolutional networks, *European Conference on Computer Vision* 2014, pp. 818–833 (Springer, Cham).
- [25] Jeff Donahue, et al., Decaf: A deep convolutional activation feature for generic visual recognition, *International Conference on Machine Learning* 2014, pp. 647–655.
- [26] X. Pan, L. Li, H. Yang, Z. Liu, J. Yang, L. Zhao, Y. Fan, Accurate segmentation of nuclei in pathological images via sparse reconstruction and deep convolutional networks, *Neurocomputing* 229 (2017) 88–99.
- [27] N. Hatipoglu, G. Bilgin, Cell segmentation in histopathological images with deep learning algorithms by utilizing spatial relationships, *Med. Biol. Eng. Comput.* (2017) 1–20.
- [28] N.H. Maerz, Image sampling techniques and requirements for automated image analysis of rock fragmentation, *Proc. FRAGBLAST* 5 (1996) 115–120.
- [29] N.H. Maerz, Aggregate sizing and shape determination using digital image processing, *Center For Aggregates Research (ICAR) Sixth Annual Symposium Proceedings* 1998, pp. 195–203.
- [30] N.H. Maerz, Online fragmentation analysis: achievements in the mining industry, *Proc., 7th Annual Symposium, ICAR*, 1999.